

This is my solution for Homework #3, which involves setting up a Linux server to manage users and directories for a small school. The goal is to configure directories for teachers and students, set appropriate permissions, and define how they can read, write, and execute files within these directories.

First, I logged into my virtual machine on the server using my credentials. Once logged in, I began my solution by creating user accounts for the staff and students at the school.

```
Ubuntu 24.04.1 Nobel Numbat (Snapshot 2) [Running]
vboxuser@Ubuntu24:~$ ssh ecuser@server-19.ec.cloud2.se
ecuser@server-19.ec.cloud2.se's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-122-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Nov  6 13:35:12 CET 2024

System load:  0.08               Processes:    113
Usage of /:   10.0% of 24.05GB   Users logged in:  0
Memory usage: 25%               IPv4 address for eth0: 134.122.51.29
Swap usage:   0%                IPv4 address for eth0: 10.18.0.33

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

*** System restart required ***
Last login: Wed Nov  6 04:01:14 2024 from 90.129.209.62
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

By using the command “sudo useradd -m <username>”, I created 14 user accounts on the server. After the accounts were created, I added comments to each user profile to specify their role (teacher or student). This helped me clearly distinguish between the users and assign appropriate permissions based on their roles.

```
ecuser@server-19:~$ sudo usermod -c "teacher" AliBoulala
ecuser@server-19:~$ sudo usermod -c "teacher" BengtSvensson
ecuser@server-19:~$ sudo usermod -c "teacher" JohnMacleod
ecuser@server-19:~$ sudo usermod -c "teacher" LottaKarlsson
ecuser@server-19:~$ sudo grep LottaKarlsson /etc/passwd
LottaKarlsson:x:1004:1004:teacher:/home/LottaKarlsson:/bin/sh
```

For the next step, I created two groups: one for teachers and one for students and assigned the appropriate users to each group. This organization allowed me to manage permissions more effectively based on the users' roles.

```
ecuser@server-19:~$ sudo gpasswd -a AliBoulala teachers
Adding user AliBoulala to group teachers
ecuser@server-19:~$ grep teachers /etc/group
teachers:x:1017:AliBoulala
```

```
ecuser@server-19:~$ grep students /etc/group
students:x:1018:AhmadArtin,HamidKhalili,HassanHaddad,JohanBokdunge,KentOlsson,LindaEk,MarcinOlofsson,RolfNordman,SalmaHabib,VincentRegalus
```

```
ecuser@server-19:~$ grep teachers /etc/group
teachers:x:1017:AliBoulala,BengtSvensson,JohnMacleod,LottaKarlsson
```

After completing the creation of the users and groups, and assigning users to their respective groups, it was time to create the directories required for each group to fulfill the assignment requirements. When a user is created in the system, a home directory is automatically created for them. However, for this step, we needed to set up shared directories for each group. To meet this requirement, I created two directories: one for the teachers and one for the students.

```
ecuser@server-19:~$ sudo mkdir teachers
ecuser@server-19:~$ ls
teachers
ecuser@server-19:~$ sudo mkdir students
ecuser@server-19:~$ ls
students  teachers
```

To set the proper permissions for accessing these directories, the first step is to change the ownership of the directories. Once the ownership is correctly set, we can then manage the permissions to control access to the directories.

```
ecuser@server-19:~$ sudo chown :students students
ecuser@server-19:~$ ls -ld students
drwxr-xr-x 2 root students 4096 Nov  6 16:13 students
ecuser@server-19:~$ sudo chown :teachers teachers
ecuser@server-19:~$ ls -ld teachers
drwxr-xr-x 2 root teachers 4096 Nov  6 16:13 teachers
```

In the final step, I used the “chmod” command to set the appropriate permissions for reading, writing, and executing files within the directories. This step was applied to both the teacher and student directories to ensure proper access control. Note that the **teacher’s** group has full access both their own group’s directory and the students' directory. On the other hand, the **students** group only has read permissions for their own directory, with no write or execute permissions.

```
ecuser@server-19:~$ sudo chmod 770 teachers
ecuser@server-19:~$ sudo chmod 447 students
```

To confirm the configuration, I used the command `ls -la` to display the users, groups, directories, and the permissions assigned to each group.

```
ecuser@server-19:~$ ls -la
total 36
drwxr-xr-x  5 ecuser ecuser 4096 Nov  6 16:13 .
drwxr-xr-x 17 root   root   4096 Nov  6 14:22 ..
-rw-----  1 ecuser ecuser   14 Nov  6 04:02 .bash_history
-rw-r--r--  1 ecuser ecuser  220 Feb 25  2020 .bash_logout
-rw-r--r--  1 ecuser ecuser 3771 Feb 25  2020 .bashrc
drwx-----  2 ecuser ecuser 4096 Oct 27 00:46 .cache
-rw-r--r--  1 ecuser ecuser    0 Oct 22 20:37 .cloud-locale-test.skip
-rw-r--r--  1 ecuser ecuser  807 Feb 25  2020 .profile
-rw-r--r--  1 ecuser ecuser    0 Nov  6 13:36 .sudo_as_admin_successful
dr--r--rwx  2 root   students 4096 Nov  6 16:13 students
drwxrwx---  2 root   teachers 4096 Nov  6 16:13 teachers
```

I considered adding passwords for each user or for each group to enhance security. However, I decided to proceed without this step, as it wasn't strictly necessary for managing the solution. That said, implementing passwords could serve as an additional layer of authentication for accessing the directories.